

# Colouring the square of a planar graph

OPG-47470

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

## ORIGINAL PROBLEM STATEMENT

Conjecture Let  $G$  be a planar graph of maximum degree  $\Delta$ . The chromatic number of its square is

- at most 7 if  $\Delta = 3$ ,
- at most  $\Delta + 5$  if  $4 \leq \Delta \leq 7$ ,
- at most  $\left\lfloor \frac{3}{2} \Delta \right\rfloor + 1$  if  $\Delta \geq 8$ .

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>